

THERMODYNAMICS AND THE ARROW OF TIME FROM THE HQIV TEMPERATURE LADDER: FINITE BLACKBODY ENTROPY, CAUSAL MONOGAMY, AND MACHINE-CHECKED DISSIPATION SIGNS

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ABSTRACT. On the shell temperature ladder $T(m) = 1/(m + 1)$ the library already packages a zeroth-law equivalence relation, a first-law finite-window balance for weighted ladder redistribution, a second-law dissipation sign on a minimal C_3 heat graph together with a CFL-bounded explicit-Euler energy witness, and a third-law-style eventual cooling below any $\varepsilon > 0$. Horizon blackbody modules define finite-shell photon number density, pressure $P = U/3$, and entropy density $s = (4/3)U/T$ without taking $\omega/T \rightarrow \infty$ limits. The *macroscopic arrow* is read as the conjunction of (i) acyclic causal access on the null lattice, (ii) nonnegative discrete entropy production, (iii) outward shell indexing toward lower ladder temperature, and (iv) opening of under-occupied early shells toward the quadratic null-shell budget (Section 7).

This is the fourth paper in the Tier-1 foundation-extension series. The immediate predecessor [4] supplies the truncated Planck spectrum and finite entropy density; the present note records what the same ladder already proves about the thermodynamic laws and the discrete arrow, and what remains interpretive. Continuum Boltzmann hierarchies and CPT microreversibility appear only as comparison vocabulary unless packaged as patch theorems. Proved items are the law-style ladder theorems, blackbody entropy positivity, the Kirchhoff detailed-balance bundle, the toy heat dissipation sign with Euler monotonicity, and the signed shell-budget ledger (the [signed shell budget](#) entry). The Jacobson–Brodie horizon accounting for $(\alpha, \gamma) = (3/5, 2/5)$ in [7] is cited as a coherence check, not a second derivation. Phase 2 opening is machine-checked in the [shell opening evolution](#) module: step-wise vertex insertion on vertex-only complexes, strict lex descent on the early/total negative budget pair, convergence to the $S3$ null reference from deficit-only horizon data, and packaging as a real Lyapunov descent plus discrete parallel Poincaré hypothesis at the converged equilibrium (build target `HQIVParallelPoincare`). The thermodynamic arrow certificate is packaged in the [thermodynamic arrow bridge](#). With link deficit identically zero, the real scaffold records shell-0 mismatch only; strict real descent holds on shell-0 opening, while the encoded natural-number lex measure strictly decreases on every non-equilibrium step. Numerical illustrations appear in Appendix B (`scripts.zip`).

Patch QFT and discrete numerics (reader contract). HQIV (Horizon-Quantized Informational Vacuum) is formulated as a *patch theory*: quantum fields, actions, and physical readouts are attached to discrete patches—shell indices $m \in \mathbb{N}$, finite spacetime index types (e.g. Fin 4), and horizon-limited charts—rather than to a hypothetical fundamental smooth spacetime obtained as a mesh-refinement or ultraviolet limit. Accordingly, when this text refers to “QFT,” it means *patch QFT*: patching rules, finite measurement ledgers, and variational identities on the discrete spine; it is not the claim that the theory is *defined* by recovering ordinary continuum quantum field theory through such a limit. The numbers that admit the sharpest statements—mass and coupling ladders, curvature ratios, shell thermodynamics, and rational witnesses in `HQIV_LEAN`—are objects of *discrete mathematics* on $\mathbb{N}$ (combinatorial growth, cumulative simplex ledgers) together with the ordered field $\mathbb{R}$ as used in the proof assistant for analysis, bounds, and phenomenological display. Smooth-manifold or continuum field notation, when it appears, is a *translation layer* for comparison with standard physics literature; it does not supersede the discrete definitions.

A continuum is not required, and in places would destroy these results. The discrete patch layer is *not* a numerical approximation to a more fundamental continuum theory awaiting future construction; it is the ontology. Where the literature treats “the continuum limit” as the goal, HQIV treats it as a *calculation approximation* for comparison with standard formulas. The continuum form is an optional convenience for comparison; the proved discrete statement is what carries the physics. In several load-bearing places—the curvature imprint $\delta_E(m)$ at shell m , the harmonic ladder masses $m_\tau \rightarrow m_\mu \rightarrow m_e$, the lock-in vev $v = \sqrt{\eta_{\text{paper}} \Omega_k(m_{\text{lockin}}; m_{\text{lockin}})}$, the proton anchor at $m = \text{referenceM} = 4$, the baryon-asymmetry $\eta \propto 6^7 \sqrt{3}/(m+1)$, and the rational coefficients $\alpha = 3/5$, $\gamma = 2/5$, $\alpha_{\text{GUT}} = 1/42$ —taking a literal $m \rightarrow \infty$ or sub-Planck continuum limit would *erase* the discrete index that is doing the predictive work. Continuum forms of these objects are therefore not preferred refinements; they are degenerate, information-losing readouts of the same patch data.

An exception: $\mathfrak{so}(8)$ as a de-facto continuum algebra from a discrete construction. Principled exception to the discrete-only stance: the closed Lie algebra $\mathfrak{so}(8) \supseteq G_2 \cup \{\Delta\}$ generated from the Fano octonion left-multiplication table and the phase-lift generator $\Delta \in (e_1, e_7)$. Although $\mathfrak{so}(8)$ is a continuous (28-dimensional real) Lie algebra, it is built here entirely from *discrete* ingredients (the seven imaginary octonion units, a finite Fano table, and one $\pi/2$ rotation), and its closure to 28 dimensions is a finite linear-algebra certificate (see the `Hqiv.S08Closure` / `Hqiv.S08ClosureInterface` stack). The Lie-group structure is therefore a *de-facto continuum algebra obtained from a discrete construction*: continuous in parameter space, but with no continuum spacetime, no continuum measure, and no UV limit attached. When this manuscript or its companions write $\exp(t\Delta) \in \text{SO}(8)$ or treat $\mathfrak{so}(8)$ rotations as smooth, those are statements internal to a finite-dimensional Lie group whose generators are certified by discrete data; they are *not* a continuum-spacetime commitment and do not require a continuum-QFT scaffolding.

1. NOMENCLATURE

$T(m)$:

Shell temperature readout $T(m) = 1/(m+1)$ on $m \in \mathbb{N}$; larger m is colder in the HQIV convention (Definition 4.1).

$\phi(m)$:

Auxiliary scalar $\phi(m) = 2(m+1)$ on the temperature ladder, shared with the O–Maxwell and HQVM sectors [6].

α, γ :

Curvature imprint and monogamy split; $\alpha = 3/5$, $\gamma = 2/5$, $\alpha + \gamma = 1$ [7, 6].

C_3 heat graph:

Minimal three-node cycle carrying the proved dissipation sign and explicit-Euler energy witness (Section 5).

Arrow readout:

Not a fifth law: the conjunction of causal acyclicity, nonnegative discrete dissipation, and outward shell indexing toward lower $T(m)$ (Section 7).

`deficitOnlyOnHorizon`:

On horizon n , every shell $m \leq n$ satisfies `shellBudgetMismatch` ≤ 0 (no excess above the quadratic budget).

`IsS3NullReference`:

Combinatorial equivalence of a vertex-only complex to `S3NullReference` n at the same horizon (alias of `IsS3NullVertexTemplate`).

shellOpeningLexPair:

Lexicographic Lyapunov pair (`totalEarlyNegativeBudget`, `totalNegativeBudget`); encoded as a single $\mathbb{N}$ for termination certificates.

2. SCOPE AND CLAIM DISCIPLINE

The HQIV programme treats horizons as information ledgers on a discrete null lattice [3, 5]. The Kirchhoff companion [4] recovers a *finite* Planck spectrum with built-in ultraviolet and infrared cutoffs; the growth companion [7] supplies a thermodynamic *coherence reading* of the same (α, γ) split that Lean forces combinatorially. **This note answers: given those inputs, what thermodynamic law statements and what arrow-of-time statements are already theorems, and what is still programme or comparison-layer physics?**

- (1) **Proved in Lean (ladder laws).** Thermal equilibrium as equality of $T(m)$; finite-window first-law balance; nonnegative entropy-production proxy on C_3 ; explicit-Euler quadratic-energy non-increase under a CFL bound; eventual shell cooling below any $\varepsilon > 0$ (the [thermodynamic laws from ladder](#) module).
- (2) **Proved in Lean (blackbody readouts).** Finite-shell energy density, $P = U/3$, $s = (4/3)U/T$, positivity of s for $T > 0$, and the Kirchhoff detailed-balance identity at each shell (the [horizon blackbody ladder](#) stack).
- (3) **Proved in Lean (signed ledger and opening, Phase 1–2).** Early-closed regime, negative budget accounting, step-wise opening with lex descent, and termination at the $S3$ null reference, packaged in the signed shell budget and shell opening entries; the conjunction of ladder laws 0 and 3 with opening convergence in the thermodynamic arrow bridge.
- (4) **Arrow packaging (partial).** Causal acyclicity and horizon-overlap monogamy remain templates in [7]; discrete heat supplies the proved C_3 dissipation sign; discrete phase evolution supplies conditional termination whenever a Lyapunov descent certificate is provided.
- (5) **Comparison layer only.** Continuum statistical mechanics, ergodicity, and textbook Boltzmann H -theorems.
- (6) **Not claimed.** Derivation of full interacting continuum thermodynamics from first principles in this note; uniqueness of all equilibrium states in field theory; microscopic time reversal as a theorem of the octonion table alone.

3. UPSTREAM SPINE CONSUMED HERE

We do not introduce new dimensionful anchors. The note consumes:

- the discrete null-lattice mode budget and $\alpha = 3/5$ imprint [5, 4];
- the rational monogamy split $\gamma = 2/5$ with $\alpha + \gamma = 1$ [7, 6];
- the auxiliary-field ladder $\phi(m) = 2(m + 1)$ threading the action sector [6];
- finite Planck means and Kirchhoff rates from [4].

Readers seeking the Jacobson-identity thermodynamic derivation of (α_d, γ_d) should use Appendix B of [7]; we cite it as a *coherence check* on the same imprint–overlap accounting, not as an independent second proof of the ladder laws proved here.

3.1. HQVM lapse and the same ϕ slot. The octonionic-action companion [6] defines the Horizon-Quantized Vacuum Metric (HQVM) lapse $N = 1 + \Phi + \phi t/c$ and identifies the *same* auxiliary ϕ with the O–Maxwell coupling and the homogeneous Friedmann branch ([lapse-dragging cross-sector identification](#) in that paper). We do not re-derive HQVM here; the thermodynamic arrow in Section 7 only needs the facts that (i) ϕ labels the temperature ladder, (ii) larger shell index means smaller $T(m)$, and (iii) the wall-clock versus apparent-age split documented in [4] selects a privileged “now” slice without adding a separate entropy functional.

3.2. Lattice origin of the temperature ladder and the rational split. The temperature ladder and the curvature-monogamy split (α, γ) are not independent inputs; both descend directly from the discrete null-lattice construction (octonionic light-cone combinatorics plus unit informational budget) developed in the light-cone and causal-growth companions [5, 7].

The octonionic left-multiplication table on the Fano plane equips every horizon with a finite mode budget whose cumulative count follows from stars-and-bars on the 3-simplex structure of the null lattice:

$$\mathcal{N}(m) = (m+2)(m+1), \quad \sum_{k=0}^n \mathcal{N}(k) = \frac{1}{3}(n+1)(n+2)(n+3).$$

In the homogeneous limit the auxiliary scalar ϕ that threads the O-Maxwell and HQVM sectors is fixed by informational-energy accounting on the universal cutout (one quadrant per octonion direction). This produces the exact linear growth

$$\phi(m) = 2(m+1)$$

in Planck units. The unique identification that renders ϕ dimensionless and consistent with the radial light-cone coordinate is the inverse-depth temperature readout

$$T(m) := \frac{1}{m+1}$$

(so that $\phi(m) = 2/T(m)$). No continuum fitting or external calibration enters; the form is the direct translation of discrete shell depth into an inverse-energy label.

The curvature-imprint coefficient α is likewise lattice-forced. The hockey-stick identity on the cumulative mode count, normalized by the quadratic multiplicity factor appearing in the denominator, yields the constant rational

$$\alpha = \frac{3}{5}$$

at *every* finite shell n (not merely in the limit). The informational-monogamy coefficient required to keep the total overlap budget equal to unity on the null lattice is then necessarily the complement

$$\gamma = 1 - \alpha = \frac{2}{5}.$$

The same rational pair is recovered from the Jacobson thermodynamic identity on horizons [9, 1], providing an independent coherence check rather than a second derivation.

Consequently the thermodynamic laws and the arrow readouts rest on the same two discrete axioms (null-lattice light-cone + unit informational budget) that underwrite the rest of the HQIV programme; the numerical values $T(m) = 1/(m+1)$ and $(\alpha, \gamma) = (3/5, 2/5)$ are outputs, not postulates.

4. THE SHELL TEMPERATURE LADDER AND EQUILIBRIUM

Definition 4.1 (Shell temperature readout). *The HQIV temperature ladder assigns to each shell index $m \in \mathbb{N}$ the dimensionless readout*

$$T(m) = \frac{1}{m+1}.$$

This is the same ordered field object used in horizon thermodynamics and epoch placement throughout the library: larger m means *colder* shells in this convention, and the map $m \mapsto T(m)$ is strictly decreasing.

Definition 4.2 (Thermal equilibrium on shells). *Shells m and n are in thermal equilibrium when $T(m) = T(n)$.*

Proposition 4.3 (Zeroth law as an equivalence relation). *Equality of the readout $T(m)$ is reflexive, symmetric, and transitive on $\mathbb{N}$.*

Proof. Machine-checked as the [zeroth-law cluster](#) in the thermodynamic laws module. $\square$

Remark 4.4 (Physical reading). *The zeroth law is not imported from continuum phenomenology: it is the statement that “same ladder temperature” is a consistent equilibrium label on the discrete index. Coupling to dimensionful MeV or kelvin displays is a downstream calibration layer (epoch modules, lock-in witnesses) and is not required for the relations in Propositions 4.3–5.5.*

5. FIRST, SECOND, AND THIRD LAWS ON THE DISCRETE SPINE

5.1. First law as finite-window redistribution. The division-algebra/zeta scaffold proves a weighted conservation identity: a finite window of shells redistributes ladder temperature so that shell weights sum to a fixed reference value T_{ref} (the [first-law finite-window balance](#)).

Theorem 5.1 (First-law style balance). *For every reference T_{ref} , window size $N > 0$, and profile parameter p ,*

$$\sum_{m < N} \text{tempLadderConserved}(T_{\text{ref}}, m) \dim \text{ShellWeight}(p, N, m) = T_{\text{ref}}.$$

This is the discrete analogue of “internal energy accounting on a finite patch”: no continuum density dU is assumed.

5.2. Second law as a sign theorem on a toy heat graph. To avoid hand-waving about dissipation, Lean carries a minimal illustrative patch—the three-node cycle C_3 —with an explicit graph Laplacian Δ (the [toy discrete heat](#) module). For any state u on the three nodes,

$$\sum_i u_i (\Delta u)_i = - \sum_i (u_i - u_{i+1})^2 \leq 0,$$

so the inner product of u with its discrete Laplacian is never positive.

Definition 5.2 (Entropy-production proxy on C_3). *Define $\Sigma(u) := - \sum_i u_i (\Delta u)_i$.*

Theorem 5.3 (Second law: nonnegative production proxy). *For all u , $\Sigma(u) \geq 0$.*

An explicit Euler update $u \mapsto u^+$ decreases the quadratic energy $\sum_i u_i^2$ provided a CFL condition $3\nu \Delta t \leq 2$ holds (the [second-law Euler monotonicity](#) theorem). This is a *discrete* irreversibility witness: it does not appeal to a continuum entropy functional whose existence is assumed.

Remark 5.4 (Scope of the toy graph). *The C_3 model is the smallest nontrivial laboratory in which the second-law sign can be exhibited and machine-checked; it is **not** a claim about 3D Navier–Stokes. Its role here is narrower: fix the sign of dissipation that the second law must inherit on a patch.*

The algebraic origin of the sign is not special to C_3 . For any undirected graph the Dirichlet form associated with the combinatorial Laplacian satisfies

$$u^T \Delta u = - \sum_{\{i,j\} \in E} (u_i - u_j)^2 \leq 0,$$

with equality if and only if u is constant on each connected component. Consequently the entropy-production proxy $\Sigma(u)$ remains nonnegative on an arbitrary finite patch (regular or irregular, sparse or dense) whose heat flow is governed by the standard graph Laplacian. Larger or topologically complex patches only modify the spectrum of relaxation eigenvalues; the sign of discrete dissipation is unchanged. The C_3 calculation therefore illustrates a general mechanism rather than a special case.

TABLE 1. Finite blackbody readouts versus large-cutoff proxy (natural units, $T = 0.05$).

m_{IR}	U	s	U/U_{5000}	s/s_{5000}
100	1708.20	45552.0	0.000343	0.000343
500	48369.5	1.289853×10^6	0.009707	0.009707
1000	196679	5.244766×10^6	0.039469	0.039469
2000	793288	2.115435×10^7	0.159196	0.159196
5000 (proxy)	4.983100×10^6	1.328827×10^8	1.000000	1.000000

5.3. Third law as eventual ladder cooling.

Theorem 5.5 (Third-law style eventual cooling). *For every $\varepsilon > 0$ there exists a shell m with $T(m) < \varepsilon$.*

Proof. Since $T(m) = 1/(m+1) \rightarrow 0$ along the index, the statement is pure archimedean order on $\mathbb{R}$; see the [third-law eventual cooling](#) theorem. $\square$

6. FINITE-MODE BLACKBODY THERMODYNAMICS

The Kirchhoff paper [4] constructs a finite Planck spectrum between an ultraviolet shell m_{UV} and an infrared cutoff m_{IR} . The horizon blackbody ladder module packages standard *readouts* on that truncated sum:

- photon number density as a finite shell sum;
- radiation pressure $P = U/3$;
- entropy density $s = (4/3)U/T$ for $T > 0$;
- detailed balance: emission equals absorption shell-wise (the [Kirchhoff detailed-balance bundle](#)).

Proposition 6.1 (Entropy density is nonnegative on the patch). *For $T > 0$ and finite cutoffs, $s \geq 0$.*

Remark 6.2 (No ultraviolet catastrophe). *Because $\mathcal{N}(m) = (m+2)(m+1)$ is finite at each shell and only finitely many shells enter a given readout, U and s are sums, not integrals requiring $\omega_{\text{max}} \rightarrow \infty$. Continuum Stefan–Boltzmann or Wien laws may be cited as calculation approximations for comparison; they are not the proved objects.*

Table 1 (reproduced by the script in Appendix B) gives a quantitative view of the approach to the continuum regime. For fixed temperature in natural units the truncated sums U and s are evaluated at increasing infrared cutoffs m_{IR} . The ratios relative to a large-cutoff proxy ($m_{\text{IR}} = 5000$) rise steadily once many shells become occupied, showing that the discrete ladder sum recovers the Stefan–Boltzmann T^4 scaling of the continuum theory in the appropriate limit (as already derived in the Kirchhoff companion).

7. THE ARROW OF TIME

Textbook thermodynamics often *defines* the arrow of time by the second law while leaving the microscopic time-reversal symmetry of Hamiltonian mechanics (Loschmidt’s paradox) untouched. In the HQIV setting the arrow is not introduced by fiat as a fifth law or as an auxiliary entropy functional. It is **derived** as the conjunction of four structures already present in the audited discrete layer:

- (1) acyclicity of the causal partial order on the null lattice (no influence from a shell’s future cone);
- (2) nonnegative discrete entropy production on any finite heat patch (the Laplacian Dirichlet form);
- (3) monotonic outward shell indexing on the temperature ladder (larger $m \Leftrightarrow$ lower $T(m)$);

- (4) forced relaxation of early under-occupied shells toward the quadratic null-shell budget via the opening evolution, with a machine-checked Lyapunov descent.

The conjunction therefore supplies an explicit, ledger-level resolution of the information-loss and Loschmidt tensions: the “missing” degrees of freedom on early closed shells are not lost but recorded as negative budget; causal acyclicity and the dissipation sign supply the statistical arrow; the ladder supplies the cosmological direction; and the opening flow supplies a constructive, non-equilibrium route from deficit-only initial data to the equilibrium reference state. Nothing in this derivation assumes a continuum entropy or CPT micro-reversibility beyond what is already encoded in the discrete poset and the Laplacian.

7.1. Causal direction on the null lattice. The growth paper [7] treats a locally finite causal partial order $\prec$ on null data. Acyclicity of influence is the operational content of “forward in time”: no shell may consume information from its own future cone without violating the poset. Horizon overlap sets $O(e_i, e_j)$ support a subadditivity template (informational monogamy): shared past/future cones cannot be double-counted without paying an overlap tax. That template is the same accounting Brodie’s thermodynamic horizon package [1] reads through Jacobson’s flux–entropy identity [9].

Lean status. Monogamy is a proved *template* on overlap functionals together with the forced split $\gamma = 1 - \alpha$; specialising it to a unique continuum entropy functional is not a theorem of this note.

7.2. Thermodynamic direction from dissipation signs. On the minimal heat patch, Theorem 5.3 fixes a direction for relaxation: states with unequal edge jumps produce strictly positive $\Sigma(u)$ unless u is a spatially constant mode on C_3 . Combined with the Euler monotonicity witness, forward numerical evolution is a *contracting* readout in quadratic energy under the stated CFL bound.

7.3. Cosmological direction: outward shell indexing. Cosmology in HQIV is not a homogeneous FLRW continuum taken as fundamental; it is a sequence of shell readouts with monotonic mode budgets. The temperature ladder makes the thermodynamic arrow coincide with **increasing shell index toward the present cold background**: larger $m \Leftrightarrow$ smaller $T(m)$. The wall-clock versus apparent-age split [4, 6] is an additional observer-sector arrow (lapse compression) orthogonal to the ladder sign but compatible with it: both select a privileged “now” slice without introducing a tunable entropy functional.

7.4. Signed shell ledger (Phase 1). The signed shell ledger (occupied vertices minus quadratic budget) lives in the null-lattice complex. Negative values on early shells ($m \leq 2$, i.e. $\xi \leq 7/2$) mark *closed* / *under-occupied* regimes; zero marks open equilibrium at the quadratic law. The normalization anchor $\xi = 7/2$ and the regime predicates (early-closed, negative budget, total negative budget) are packaged in the [signed shell budget](#) entry of the Lean index.

7.5. Discrete opening evolution (Phase 2). Step and equilibrium. On vertex-only complexes, the opening step inserts one null-shell vertex on the smallest active shell carrying negative budget, when any exists; otherwise the step is none. The flow (packaged as shell opening evolution) fixes the state at equilibrium (total negative budget zero, equivalently the identity step at equilibrium).

Theorem 7.1 (Strict lex descent and termination). *For every coupling $\alpha > 0$, horizon index n , and reference `href` with $0 < K n \alpha$, off equilibrium the opening evolution strictly decreases the lexicographic pair $(\text{totalEarlyNegativeBudget}, \text{totalNegativeBudget})$ on each some step; hence `FlowTerminatesAt` and `totalNegativeBudget` $M' = 0$ at the reached state.*

Proof. Machine-checked in the shell opening evolution module (see the Lean index). □

Theorem 7.2 (Deficit-only horizon $\Rightarrow$ S3NullReference). *Let M_0 be vertex-only with $\text{maxVertexShell } M_0 = n$ and $\text{deficitOnlyOnHorizon } M_0 n$. Then there exist k and M' such that opening iterates M_0 to M' and $\text{IsS3NullReference } M' n$.*

Proof. The chain is: zero total negative budget at the terminal state; no negative shells on the active range; quadratic null-shell growth on the horizon from the deficit-only predicate; then the quadratic growth identity implies the S3 null reference. See the shell opening module in the Lean index. $\square$

Remark 7.3 (Two Lyapunov layers). *With link deficit identically zero, the real-valued Lyapunov functional records shell-0 mismatch only; it is non-increasing on opening steps but strictly decreases only when shell-0 opens. The proved arrow certificate for parallel Poincaré uses the encoded natural-number lex measure, not unconditional real descent on every shell index.*

7.6. Thermodynamic and parallel-Poincaré packaging.

Definition 7.4 (Thermodynamic arrow certificate). *A $\text{ThermodynamicArrowFromShellOpening } n$ packages (i) a positive curvature-channel reference, (ii) the zeroth-law reflexivity witness on $T(m)$, (iii) third-law eventual cooling, and (iv) opening convergence to the S3 null reference for every deficit-only horizon initial state at n .*

Proposition 7.5 (Default witness at $\alpha = 1$). *For each n and positive reference with $0 < K n \cdot 1$, the structure $\text{thermodynamicArrowFromShellOpening}$ is inhabited.*

Theorem 7.6 (Discrete parallel Poincaré at the converged horizon). *Under the hypotheses of Theorem 7.2, there exist k , M' , and a $\text{DiscreteParallelPoincareHypothesis } H$ with $H.\text{evo}$ the opening evolution, $H.\text{data}.M = M'$, and the S3 null reference at $H.\text{data}.M$.*

Proof. The packaged witness in the thermodynamic arrow bridge, via the real-descent constructor in the parallel Poincaré scaffold (see Lean index). $\square$

Template track (reference). Independently, template-pinned evolution at the S3 null reference already satisfies the discrete parallel Poincaré property on the certificate track without opening dynamics (see the reference model entry in the Lean index).

Physical readout. Negative early shell-budget mismatch is a concentrated low-occupancy ledger state; opening is forced insertion toward the quadratic law on a finite horizon—not a negative curvature imprint (the curvature-density terms remain positive in the HQIV convention). The conjunction of Theorem 7.2 with the C_3 dissipation sign and Proposition 5.5 supplies the discrete arrow readout advertised in the abstract: causal direction from upstream growth, thermodynamic direction from dissipation, cosmological direction from the ladder, and combinatorial relaxation toward the null-shell reference template. In this sense the four-part conjunction is a *derivation* of the arrow from structures already present in the discrete layer, not an additional definition; it directly exhibits how information that would be “lost” in a continuum description is retained in the signed budget ledger and how microscopic reversibility on individual edges is compatible with macroscopic directionality once the global causal poset and the Lyapunov descent are taken into account.

8. HOW THE PIECES COMPOSE

Structure	Thermodynamic role	Arrow role
C_3 heat sign	second law proxy	relaxation direction
Finite Planck sum	U, P, s on patch	no UV catastrophe
$T(m) = 1/(m+1)$	zeroth law label	m increases $\Rightarrow$ cooler
Causal $\prec$ + overlaps	Jacobson/Brodie coherence	no future influence
α, γ split	horizon accounting	fixed monogamy budget
Signed shell-budget mismatch	budget ledger	early opening driver
Shell opening step	mode insertion	Thm. 7.2

9. OUTLOOK AND HONEST GAPS

A short note on related discrete approaches may be useful for context. The present derivation shares with causal-set theory (Sorkin and collaborators) the use of a discrete partial order and a fundamental length scale, but replaces the continuum sprinkling measure with an explicit octonionic null-lattice mode budget and supplies machine-checked signs for dissipation and Lyapunov descent rather than statistical assumptions. It is closer in spirit to graph-based or finitary approaches to thermodynamics (e.g., discrete Laplacian heat flow on irregular networks) than to cellular-automaton or 't Hooft-style deterministic hidden-variable models, since the arrow here is carried by the combination of a proved causal poset, a non-positive Dirichlet form, and an explicit opening evolution rather than by an underlying deterministic micro-dynamics. A fuller comparison lies outside the scope of this note.

Natural extensions, stated as programme items rather than hidden assumptions:

- (1) Shell-resolved heat equation on the full null lattice (lifting the C_3 dissipation sign and Euler witness to the complete adjacency structure of the discrete light-cone); a follow-up note on this programme item is planned.
- (2) A counting entropy $S = \log \Omega_{\text{accessible}}$ from octonionic mode budgets on shells.
- (3) Package monogamy subadditivity as a single theorem specialised to HQIV readouts K and $\mathcal{R}$.
- (4) Connect the third-law cooling limit to CMB and BBN epoch witnesses [2] as comparison-layer temperature displays.
- (5) **Big-Bang readout (discrete opening track, proved).** Theorem 7.2 and Theorem 7.6 formalise opening from deficit-only horizon data to the $S3$ null reference and a packaged discrete parallel Poincaré hypothesis; continuum singularity or bounce narratives remain comparison-layer vocabulary only.
- (6) **Topology gaps (unchanged).** The $\chi = 0$ combinatorial-sphere certificate is obstructed on the vertex-only $S3$ null reference; nontrivial link deficit and shell-resolved real Lyapunov on all indices are not discharged in the opening certificate.

APPENDIX A. LEAN MODULE INDEX

The Lean development lives in the public slimmed mirror <https://github.com/HQIV/hqiv-lean> [8]. All theorems cited in the body are sorry-free at the time of writing. Each entry below carries a reader-friendly anchor (used by hyperlinks in the body), a one-line description, and clickable links to the Lean module and identifiers.

Thermodynamic laws from ladder.:

Zeroth–third law packaging on the temperature ladder (equivalence, finite-window balance, C_3 dissipation sign, Euler monotonicity, eventual cooling). Lean: `zerothLaw\_refl, firstLaw\_tempLadder\_dimShellWeight, secondLaw\_entropyProduction\_nonneg, secondLaw\_euler\_step\_energy\_nonincreasing, thirdLaw\_eventually\_below` in [Hqiv/Physics/ThermodynamicLawsFromLadder.lean](#).

Zeroth law cluster.:

thermalEquilibrium, zerothLaw_symm, zerothLaw_trans in [Hqiv/Physics/ThermodynamicLawsFromLadder.lean](#).

First-law finite-window balance.:

firstLaw_tempLadder_dimShellWeight (imports [Hqiv/Physics/DivisionAlgebraZetaScaffold.lean](#)).

Toy discrete heat on C_3 .:

Laplacian sign and entropy-production proxy. Lean: `sum\_u\_laplacianCycle3\_eq\_neg\_jump\_sq`, `eulerHeatStep3\_sum\_sq\_le\_sum\_sq\_of\_three\_mul\_dt\_nu\_le\_two` in [Hqiv/Physics/ToyDiscreteHeat.lean](#).

Second-law Euler monotonicity.:

secondLaw_euler_step_energy_nonincreasing in [Hqiv/Physics/ThermodynamicLawsFromLadder.lean](#).

Third-law eventual cooling.:

thirdLaw_eventually_below in [Hqiv/Physics/ThermodynamicLawsFromLadder.lean](#).

Horizon blackbody ladder readouts.:

Finite-shell U , $P = U/3$, $s = (4/3)U/T$, positivity, and Kirchhoff detailed-balance bundle. Lean: `entropyDensity`, `entropyDensity\_nonneg`, `T\_times\_entropy\_eq` and the Kirchhoff bundle in [Hqiv/Physics/HorizonBlackbodyLadder.lean](#).

Horizon blackbody spectrum.:

Finite Planck means and patch ontology. [Hqiv/Physics/HorizonBlackbodySpectrum.lean](#).

Kirchhoff detailed-balance bundle.:

[Hqiv/Physics/HorizonBlackbodyLadder.lean](#) (§4).

Discrete phase evolution scaffold.:

Lyapunov and termination infrastructure (used by opening flow). [Hqiv/Topology/DiscretePhaseEvolution.lean](#).

Signed shell budget (Phase 1).:

Early-closed regime, negative budget, and S_3 null reference predicates on the null-lattice complex. Lean: `shellXi`, `isEarlyClosedShell`, `negativeBudget`, `totalNegativeBudget` and reference identities in [Hqiv/Topology/SignedShellBudget.lean](#).

IsS3NullReference bridge.:

Combinatorial null reference and quadratic growth identities. [Hqiv/Topology/DiscreteNullLatticeComplex.lean](#).

Shell opening evolution (Phase 2).:

`shellOpeningStep`, lex Lyapunov descent, termination at `IsS3NullReference`, and deficit-only convergence. Lean: `shellOpeningStep`, `shellOpeningRealLyapunovDescent`, `shellOpeningStep\_reaches\_S3NullReference`, `deficitOnlyOnHorizon\_of\_shellOpening\_iterate` in [Hqiv/Topology/ShellOpeningEvolution.lean](#).

Parallel Poincaré scaffold.:

[Hqiv/Topology/ParallelPoincareScaffold.lean](#).

Template-pinned reference model.:

[Hqiv/Topology/ParallelPoincareReferenceModel.lean](#).

Build target HQIVParallelPoincare.:

Aggregates scaffold, reference model, shell opening, and thermodynamic bridge (lake build `HQIVParallelPoincare`).

Thermodynamic arrow bridge.:

Packages the full arrow certificate (zeroth+third laws + opening convergence). [Hqiv/Physics/ThermodynamicArrowFromShellOpening.lean](#).

Shell temperature readout.:

Hqiv.T, T_eq in [Hqiv/Geometry/AuxiliaryField.lean](#).

Forced $\alpha = 3/5$, $\gamma = 2/5$.:

alpha_gamma_forced_pair in [Hqiv/Geometry/AlphaGammaForcedByLattice.lean](#).

Lapse dragging (HQVM / O–Maxwell cross-sector).:

Documented in [6]; Lean: HQVM_lapse, same_phi_in_0_Maxwell_and_HQVM in [Hqiv/Geometry/HQVMetric.lean](#) and [Hqiv/Physics/GRFromMaxwell.lean](#).

APPENDIX B. REPRODUCER SCRIPTS

The script `thermo_ladder_and_c3_heat.py` in the supplementary `scripts.zip` bundle checks the ladder values $T(m)$, exhibits a nonnegative entropy-production proxy on C_3 , and confirms Euler-step energy non-increase under the CFL bound stated in Section 5. It also generates the finite-versus-large-cutoff blackbody comparison data reproduced in Table 1 (fixed $T = 0.05$ in natural units, UV cutoff zero). Running the script with the `--json` flag emits the full numerical record in machine-readable form. See `papers/thermodynamics_arrow/scripts/README.md` in the HQIV-LEAN repository [8] for the exact reproduction command and licensing.

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¹AI-assisted drafting and repository tooling were used during manuscript preparation; formal claims refer to the cited Lean modules.